

Number Sense

Module 1 Test

Grade 9 • MTH1W

Time: 50 minutes • Total: 35 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Answers only earn the answer mark. Marks for each question are shown in brackets.

Section A — Knowledge & Understanding

1. Evaluate: $(-15) + (-6) - (-4)$. **[2]**

2. Evaluate: $30 - 4 \times 2^3 \div 8$. **[2]**

3. Simplify, leaving your answer as a single power: $6^5 \times 6^2 \div 6^3$. **[2]**

4. Evaluate in lowest terms: $\frac{3}{4} - \frac{5}{8}$. **[2]**

5. Write $\frac{5}{8}$ as a decimal and as a percent. **[2]**

6. State whether $\sqrt{44}$ is rational or irrational, and name the two consecutive whole numbers it lies between. **[2]**

7. Find 30% of 250. **[2]**

8. A car travels 180 km in 3 hours at a steady speed. Find its speed as a unit rate. **[2]**

Section B — Application

9. Evaluate, showing each step, in lowest terms: $(\frac{2}{3} + \frac{1}{2}) \times \frac{6}{5}$. **[3]**
10. A phone costs \$540 before tax. Find the total cost with 13% HST, and state the amount of tax paid. **[3]**
11. Simplify to a single power, then evaluate: $(5^2)^2 \div 5^3 \times 5^0$. **[2]**
12. A fruit punch mixes orange juice to pineapple juice in the ratio 3 : 5. To make 4000 mL of punch, how much of each juice is needed? **[3]**

Section C — Thinking

13. A jacket's price is first increased by 20%. Later the new price is reduced by 20%. After both changes the jacket sells for \$96. Find the original price. Then decide, with reasoning, whether the final price equals the original price. **[5]**

Section D — Communication

14. Jordan claims that $\sqrt{16} + \sqrt{9} = \sqrt{25}$. Is Jordan correct? Justify your answer with a calculation and a clear explanation. **[3]**

— END OF EXAM —